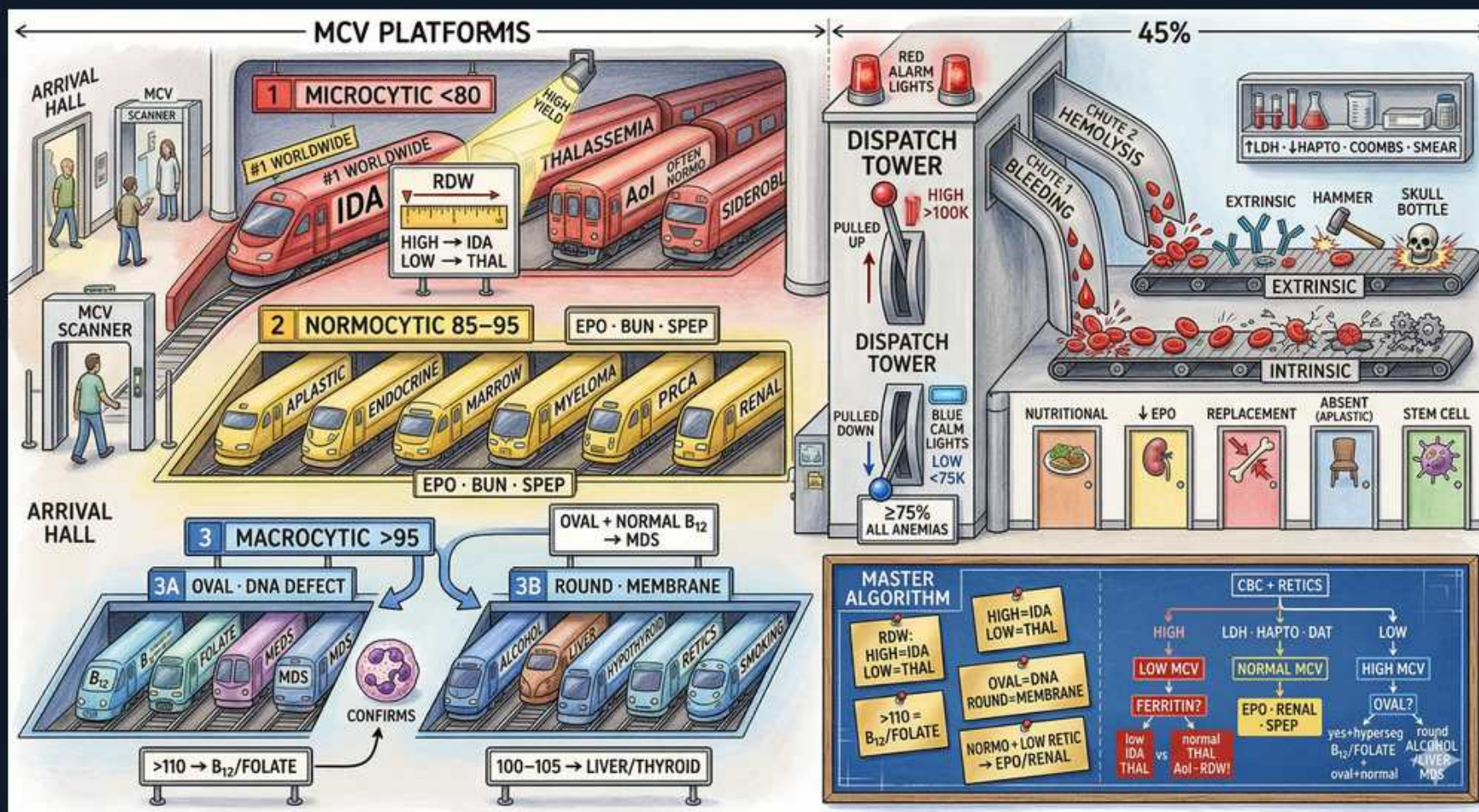


Anemia — MCV Classification & Master Algorithm



1. Microcytic <80

WHAT YOU SEE

The red '1 MICROCYTIC <80' platform where a train of tiny red boxcars (small RBCs) is parked — the smallest train on the smallest platform = microcytic anemia, MCV under 80.

WHAT IT ENCODES

MCV <80 fL = microcytic anemia, reflecting defective hemoglobin synthesis. The classic differential is captured by 'TAILS': Thalassemia, Anemia of chronic disease (can be normo- or microcytic), Iron deficiency, Lead poisoning, Sideroblastic anemia. First-line workup is iron studies (ferritin, serum iron, TIBC, transferrin saturation) plus RDW.

BOARD TRAP

WRONG answer: assuming every microcytic anemia is iron deficiency and starting iron empirically. Discriminator: check ferritin first — thalassemia, lead, and sideroblastic anemia have normal/high iron stores, and giving iron to a thalassemia trait patient risks iron overload.

2. IDA most common

WHAT YOU SEE

The IDA boxcar tagged '#1 WORLDWIDE' with 'RDW HIGH' beside it — labelled as the single most common anemia on the planet, with a wide RDW marking the mixed cell sizes of iron deficiency.

WHAT IT ENCODES

Iron deficiency is the most common anemia worldwide. Pattern: LOW ferritin (most specific test), low serum iron, HIGH TIBC/transferrin, low transferrin saturation (<15%), and HIGH RDW (anisocytosis as new microcytes mix with older normocytes). In adult men or postmenopausal women, IDA mandates a GI source workup (colonoscopy) for occult bleeding.

BOARD TRAP

WRONG answer: stopping at 'low ferritin = treat with iron' without finding the cause in an at-risk adult. Discriminator: unexplained IDA in a man or postmenopausal woman is GI malignancy until proven otherwise — pursue endoscopy/colonoscopy, don't just replace iron.

3. Thalassemia low RDW

WHAT YOU SEE

The THALASSEMIA boxcar marked with a high RBC count and a low/normal RDW — many uniformly tiny cars (high RBC, narrow size spread) flags thalassemia rather than iron deficiency.

WHAT IT ENCODES

Thalassemia trait shows a NORMAL or HIGH RBC count with disproportionately low MCV, a NORMAL/low RDW (uniform small cells), and normal/high ferritin. The Mentzer index (MCV/RBC) <13 favors thalassemia; >13 favors IDA. Confirm beta-thalassemia with Hb electrophoresis (↑HbA2); alpha-thal often needs genetic testing.

BOARD TRAP

WRONG answer: diagnosing iron deficiency in a microcytic patient with a NORMAL RDW and high RBC count. Discriminator: normal RDW + high RBC + normal ferritin = thalassemia trait, not IDA. A normal HbA2 does NOT exclude alpha-thalassemia (need genetic studies).

4. Anemia of chronic dz

WHAT YOU SEE

The ANEMIA-OF-CHRONIC-DISEASE car straddling the line between the microcytic and normocytic platforms — drawn on the border because ACD is usually normocytic but can slip into microcytic.

WHAT IT ENCODES

Anemia of chronic disease/inflammation is usually normocytic but can be mildly microcytic. Hepcidin (up-regulated by inflammation) traps iron in macrophages: LOW serum iron, LOW TIBC, and HIGH/normal ferritin (acute-phase reactant). It sits on the micro/normocytic border.

BOARD TRAP

WRONG answer: calling it iron deficiency because serum iron is low. Discriminator: in ACD the TIBC is LOW and ferritin is HIGH/normal, whereas IDA has HIGH TIBC and LOW ferritin. When ACD and IDA coexist, a low transferrin saturation with soluble transferrin receptor or a bone marrow iron stain settles it.

5. Normocytic 85-95

WHAT YOU SEE

The orange '2 NORMOCYTIC 85-95' platform with a train of normal-sized cars — the middle platform for middle-sized cells, MCV 80-100.

WHAT IT ENCODES

MCV 80–100 fL = normocytic anemia. The pivotal next step is the RETICULOCYTE count/index, which splits causes into hypoproliferative (low retic = marrow not responding) vs hyperproliferative (high retic = blood loss or hemolysis).

BOARD TRAP

WRONG answer: ordering iron studies/B12 first for a normocytic anemia. Discriminator: the retic count is the branch point in normocytic anemia — low retic points to marrow/renal/chronic-disease causes; high retic points to bleeding or hemolysis.

6. EPO/BUN/SPEP

WHAT YOU SEE

A workup checklist reading 'EPO • BUN • SPEP' on the normocytic platform — the labs you order when retics are low: kidney/EPO for renal anemia, SPEP for myeloma.

WHAT IT ENCODES

Hypoproliferative (low-retic) normocytic workup: BUN/creatinine and EPO level (CKD → inadequate erythropoietin), SPEP/free light chains (multiple myeloma), TSH (hypothyroidism), and consider marrow exam. Anemia of chronic disease and renal anemia are the most common causes.

BOARD TRAP

WRONG answer: ignoring renal function in a normocytic anemia with low retic. Discriminator: CKD-related EPO deficiency is a leading, easily missed cause — check creatinine; rouleaux/elevated total protein should trigger SPEP for myeloma.

7. Aplastic / marrow

WHAT YOU SEE

The low-retic marrow-failure cluster of cars labelled APLASTIC, MARROW infiltration, MM (myeloma) and RENAL — the empty/broken-train causes where the marrow itself is the problem.

WHAT IT ENCODES

Low-retic normocytic causes from the marrow itself: aplastic anemia (pancytopenia, hypocellular marrow), marrow infiltration/myelophthisis (fibrosis, mets, leukemia — gives leukoerythroblastic smear with teardrops and nucleated RBCs), multiple myeloma, and pure red cell aplasia (e.g., parvovirus B19, especially in immunocompromised/sickle cell aplastic crisis).

BOARD TRAP

WRONG answer: attributing pancytopenia with a 'dry tap' purely to aplastic anemia. Discriminator: a dry/blood tap with teardrop cells and nucleated RBCs suggests marrow infiltration or fibrosis (myelophthisis), which needs a biopsy — aplastic anemia is hypocellular but aspirable, not fibrotic.

8. Macrocytic >95

WHAT YOU SEE

The blue '3 MACROCYTIC >95' platform with a train of oversized blue cars — the biggest train on the biggest platform = macrocytic anemia, MCV over 100.

WHAT IT ENCODES

MCV >100 fL = macrocytic anemia. The first split is megaloblastic (impaired DNA synthesis → oval macrocytes + hypersegmented neutrophils) vs non-megaloblastic (round macrocytes, normal neutrophils). MCV >110–120 strongly favors megaloblastic.

BOARD TRAP

WRONG answer: treating all macrocytosis as B12/folate deficiency. Discriminator: the smear and degree of macrocytosis separate them — hypersegmented neutrophils + oval macrocytes + very high MCV = megaloblastic; round macrocytes without hypersegmentation = liver/alcohol/thyroid or reticulocytosis.

9. Oval / DNA defect

WHAT YOU SEE

The '3A OVAL — DNA DEFECT' branch of egg-shaped (oval) macrocyte cars, with '>110 → B12/FOLATE' — oval cells plus a very high MCV point to a DNA-synthesis (B12/folate) defect.

WHAT IT ENCODES

Megaloblastic macrocytosis (oval macrocytes, hypersegmented neutrophils, MCV often >110): B12 deficiency, folate deficiency, MDS, and drugs blocking DNA synthesis (methotrexate, hydroxyurea, zidovudine, phenytoin, trimethoprim). Both B12 and folate deficiency raise homocysteine, but only B12 deficiency raises methylmalonic acid (MMA).

BOARD TRAP

WRONG answer: giving folate to a patient who actually has B12 deficiency. Folate corrects the anemia/MCV but the neurologic damage (subacute combined degeneration — dorsal columns, paresthesias, ataxia) progresses irreversibly. Discriminator: always check/replace B12 before folate; elevated MMA confirms B12 (not folate) deficiency.

10. Round / membrane

WHAT YOU SEE

The '3B ROUND — MEMBRANE' branch of round macrocyte cars, with '100-105 → LIVER/THYROID' — round (not oval) macrocytes and only mild enlargement point to liver/alcohol/thyroid.

WHAT IT ENCODES

Non-megaloblastic (round) macrocytosis, usually mild (MCV ~100–110): liver disease, alcohol (direct marrow toxicity, can occur without B12/folate deficiency), hypothyroidism, MDS, and reticulocytosis (large young RBCs) from hemolysis/bleeding. No hypersegmented neutrophils.

BOARD TRAP

WRONG answer: working up B12/folate exhaustively while missing a high reticulocyte count. Discriminator: brisk reticulocytosis (hemolysis or recent bleed) raises MCV because reticulocytes are large — check the retic count before assuming a nutritional megaloblastic cause.

11. Hemolysis ↑retic

WHAT YOU SEE

The HEMOLYSIS chute on the high-retic side with its 'LDH ↑, hapto ↓, Coombs, smear' checklist — the destruction track that triggers the hemolysis panel.

WHAT IT ENCODES

Hemolysis labs: HIGH LDH, HIGH reticulocytes, HIGH indirect (unconjugated) bilirubin, and LOW haptoglobin (best single marker for intravascular hemolysis). Then split immune vs non-immune with the direct antiglobulin (Coombs) test, and read the smear (schistocytes → MAHA; spherocytes → HS/AIHA; bite cells → G6PD).

BOARD TRAP

WRONG answer: calling a Coombs-negative hemolytic anemia 'autoimmune.' Discriminator: a NEGATIVE direct Coombs points away from AIHA toward intrinsic/mechanical causes (G6PD, hereditary spherocytosis, PNH, MAHA, prosthetic valve) — let the smear and Coombs direct the workup.

12. Bleeding ↑retic

WHAT YOU SEE

The acute-BLEEDING car that tripped the red 'HIGH >100K' alarm in the dispatch tower — brisk blood loss spikes the reticulocyte alarm even before the hematocrit falls.

WHAT IT ENCODES

Hyperproliferative (high-retic) normocytic anemia = acute blood loss OR hemolysis. Acute hemorrhage may show a NORMAL Hb/Hct in the first hours before plasma refill, then a falling hematocrit with reticulocytosis appearing at 2–3 days.

BOARD TRAP

WRONG answer: excluding significant acute hemorrhage because the initial hemoglobin is normal. Discriminator: in early acute blood loss the Hb/Hct lags (whole-blood loss, no time to hemodilute) — trend the hematocrit and watch vitals/orthostatics rather than relying on a single early value.

13. Ferritin low = IDA

WHAT YOU SEE

The MASTER-ALGORITHM line reading 'low ferritin → IDA' — a single low-ferritin reading at the bottom of the flowchart routes straight to iron deficiency.

WHAT IT ENCODES

Ferritin is the most useful and most SPECIFIC test for iron deficiency: a low ferritin (<15–30 ng/mL) essentially confirms IDA. Caveat: ferritin is an acute-phase reactant, so in inflammation it can be falsely normal/high even with true iron deficiency (then look at transferrin saturation/soluble transferrin receptor).

BOARD TRAP

WRONG answer: ruling out iron deficiency because ferritin is 'normal' in an inflamed/hospitalized patient. Discriminator: with coexisting inflammation, a ferritin up to ~100 ng/mL can still represent iron deficiency — use transferrin saturation, sTfR, or marrow iron when ACD and IDA may coexist.

14. Master algorithm

WHAT YOU SEE

The blue 'MASTER ALGORITHM' control panel in the corner, wiring MCV → cause for every branch — the master flowchart that ties all three platforms together.

WHAT IT ENCODES

The unifying approach: (1) confirm anemia and bin by MCV (micro <80 / normo 80–100 / macro >100); (2) microcytic → iron studies + RDW (+ Mentzer index, Hb electrophoresis); (3) normocytic → reticulocyte count (low = marrow/renal/ACD; high = bleed/hemolysis); (4) macrocytic → smear for megaloblastic features (B12/MMA, folate) vs non-megaloblastic (liver/alcohol/thyroid/retic).

BOARD TRAP

WRONG answer: relying on MCV bins alone and ignoring RDW and reticulocytes. Discriminator: MCV starts the algorithm but RDW (anisocytosis) and the retic count are what actually separate IDA vs thalassemia and hypo- vs hyper-proliferative causes — mixed deficiencies (e.g., IDA + B12) can give a deceptively normal MCV with a high RDW.

15. Extrinsic vs intrinsic

WHAT YOU SEE

On the hemolysis (high-retic) side, an EXTRINSIC bay drawn with a hammer/wrench (outside forces breaking the cell) split from an INTRINSIC bay (the defect is inside the RBC) — the two ways a red cell can be destroyed.

WHAT IT ENCODES

Hemolysis splits into EXTRINSIC (defect outside the RBC — immune AIHA via Coombs, microangiopathy/MAHA with schistocytes, mechanical valve, infections like malaria/babesia, hypersplenism) versus INTRINSIC (defect within the RBC — membrane: hereditary spherocytosis/elliptocytosis; enzyme: G6PD/pyruvate kinase; hemoglobin: sickle/thalassemia; PNH). The direct Coombs (DAT) is the pivotal first split — positive = immune/extrinsic, negative = intrinsic or mechanical. Intravascular hemolysis (low haptoglobin, hemoglobinuria, very high LDH) vs extravascular (spherocytes, splenomegaly) further localizes it.

BOARD TRAP

WRONG answer: assuming a Coombs-negative hemolytic anemia must still be autoimmune and starting steroids. Discriminator: a NEGATIVE DAT redirects you to intrinsic/mechanical causes (G6PD bite cells, hereditary spherocytosis, PNH flow cytometry, MAHA schistocytes, prosthetic valve) — let the smear and DAT pick the bay before treating.

16. Stem cell / PNH

WHAT YOU SEE

The 'STEM CELL' bay on the low-retic marrow-failure side with a damaged stem-cell icon — the root-level failure where the hematopoietic stem cell itself is the problem (aplastic anemia, MDS, PNH).

WHAT IT ENCODES

Stem-cell-level disease produces hypoproliferative (low-retic) anemia, often with pancytopenia: aplastic anemia (immune-mediated stem cell destruction → hypocellular marrow), MDS (clonal dysplasia, often macrocytic with cytopenias), and paroxysmal nocturnal hemoglobinuria (PIGA mutation → loss of GPI-anchored CD55/CD59 → complement-mediated intravascular hemolysis, cytopenias, and thrombosis). PNH is diagnosed by flow cytometry (absent CD55/CD59); treat with the anti-C5 agent eculizumab/ravulizumab.

BOARD TRAP

WRONG answer: working up unexplained pancytopenia or thrombosis at an unusual site without considering PNH. Discriminator: aplastic anemia, PNH, and MDS overlap and coexist — PNH presents with intravascular hemolysis (hemoglobinuria, low haptoglobin, high LDH) PLUS thrombosis and cytopenias; confirm with flow cytometry (CD55/CD59), not Ham's test.